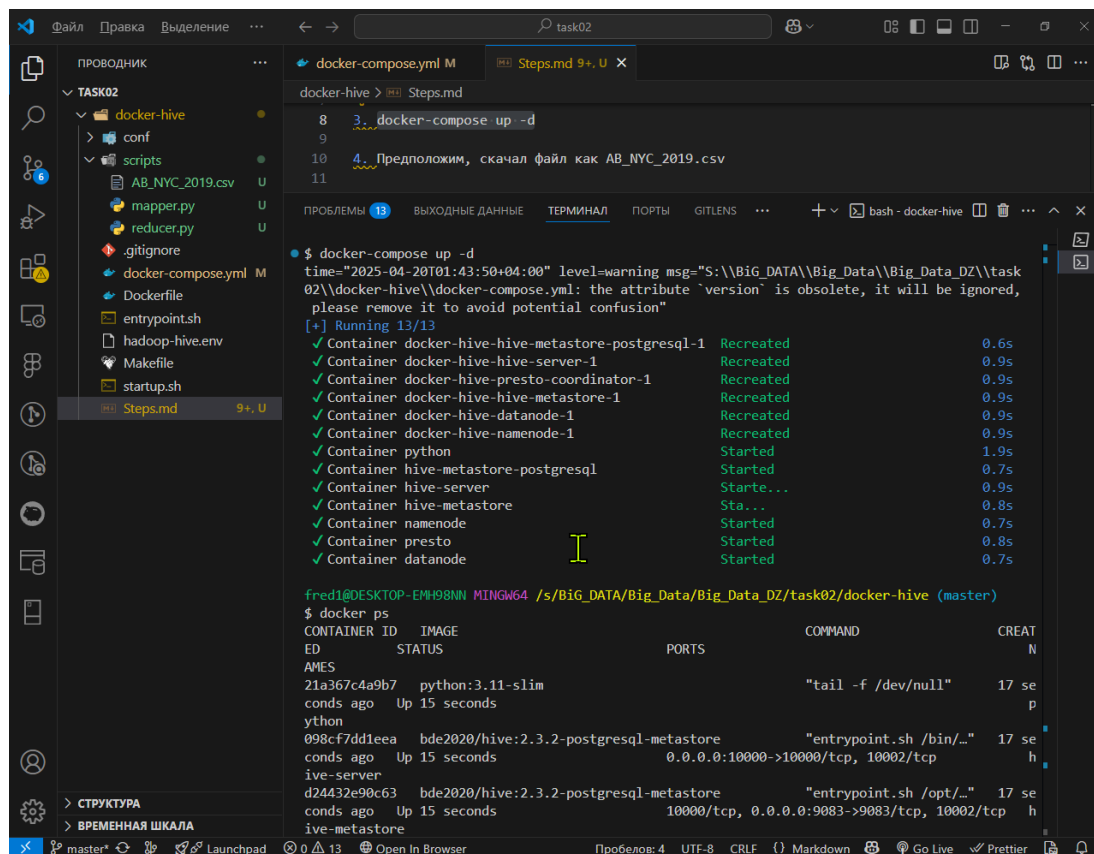
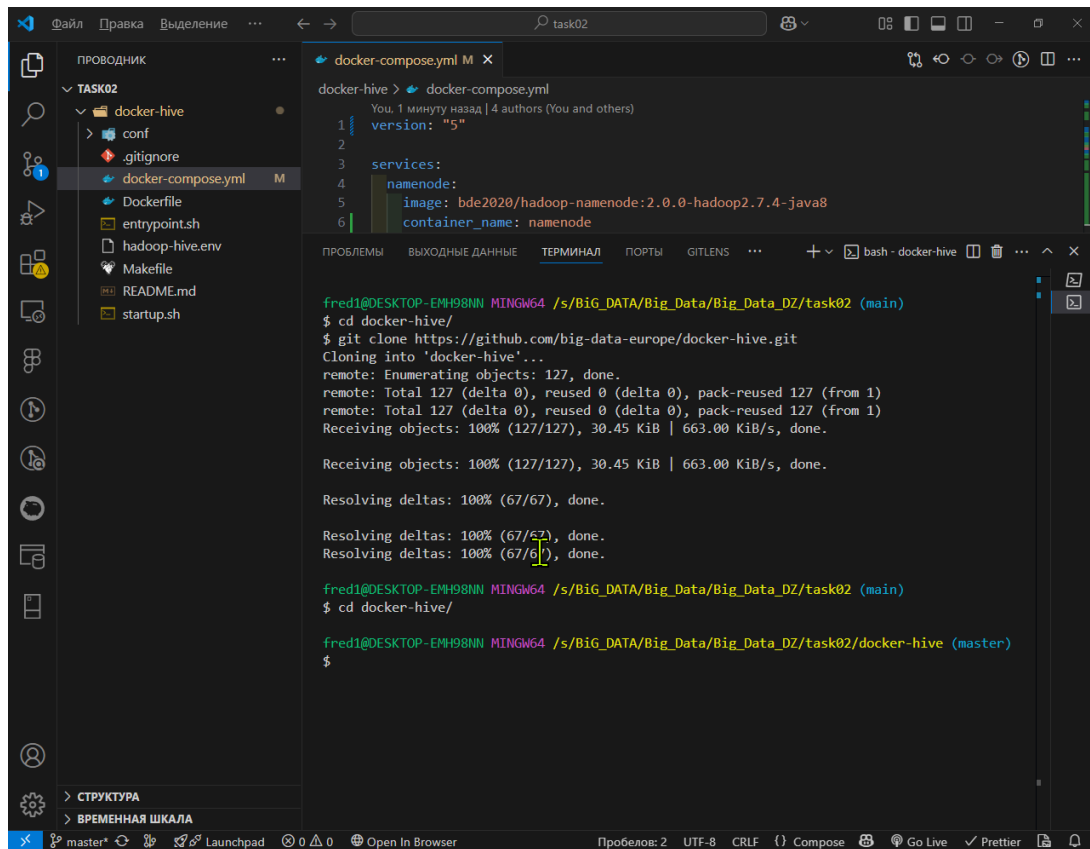




```
Файл  Правка  Выделение  ...  task02
docker-compose.yml M  Steps.md 9+, U
docker-hive > Steps.md
10 4. Предположим, скачал файл как AB_NYC_2019.csv
11
12 * docker cp AB_NYC_2019.csv namenode:/tmp/
13 * docker cp AB_NYC_2019.csv docker-hive-namenode-1:/tmp/
14

ПРОБЛЕМЫ 13 ВЫХОДНЫЕ ДАННЫЕ ТЕРМИНАЛ ПОРТЫ GITLENS QUERY RESULTS (PREVIEW)
+ bash - docker-hive

$ docker ps
CONTAINER ID IMAGE PORTS COMMAND CREATED STATUS
21a367c4a9b7 python:3.11-slim "tail -f /dev/null" 17 seconds ago Up 15 seconds
098cf7dd1eea bde2020/hive:2.3.2-postgresql-metastore "entrypoint.sh /bin/..." 17 seconds ago Up 15 seconds
d24432e90c63 bde2020/hive:2.3.2-postgresql-metastore "entrypoint.sh /opt/..." 17 seconds ago Up 15 seconds
f43207fb3209 bde2020/hadoop-namenode:2.0.0-hadoop2.7.4-java8 "entrypoint.sh /run/..." 17 seconds ago Up 15 seconds (hea
lth: starting) 0.0.0.0:50070->50070/tcp namenode
4324ab4adb8b shawnzhu/prestodb:0.181 "/bin/launcher run" 17 seconds ago Up 15 seconds
730dc23fe7ae bde2020/hadoop-datanode:2.0.0-hadoop2.7.4-java8 "entrypoint.sh /run/..." 17 seconds ago Up 15 seconds (hea
lth: starting) 0.0.0.0:50075->50075/tcp datanode
c5ca1bb13c0e bde2020/hive-metastore-postgresql:2.3.0 "/docker-entrypoint..." 17 seconds ago Up 15 seconds
5432/tcp hive-metastore-postgresql

fred1@DESKTOP-EMH98NN MINGW64 /s/Big_DATA/Big_Data/Big_Data_DZ/task02/docker-hive (master)
$ docker cp AB_NYC_2019.csv namenode:/tmp/
CreateFile S:\Big_DATA\Big_Data\Big_Data_DZ\task02\docker-hive\AB_NYC_2019.csv: The system cannot find the file specified.

fred1@DESKTOP-EMH98NN MINGW64 /s/Big_DATA/Big_Data/Big_Data_DZ/task02/docker-hive (master)
$ docker cp scripts/AB_NYC_2019.csv namenode:/tmp/
Successfully copied 7.08MB to namenode:/tmp/

fred1@DESKTOP-EMH98NN MINGW64 /s/Big_DATA/Big_Data/Big_Data_DZ/task02/docker-hive (master)
$
```

```
Файл  Правка  Выделение  ...  task02
docker-compose.yml M  Steps.md 9+, U
docker-hive > Steps.md
17 * 1. docker exec -it namenode bash
18 * 2. docker exec -it docker-hive-namenode-1 bash
19 * 1. hdfs dfs -mkdir -p /user/hive/warehouse/airbnb
20 * 2. docker exec -it docker-hive-namenode-1 hdfs dfs -mkdir -p /data/airbnb
21 * 1. hdfs dfs -put /tmp/AB_NYC_2019.csv /user/hive/warehouse/airbnb/
22 * 2. docker exec -it docker-hive-namenode-1 hdfs dfs -put /tmp/AB_NYC_2019.csv /data/airbnb/
23 * exit

ПРОБЛЕМЫ 13 ВЫХОДНЫЕ ДАННЫЕ ТЕРМИНАЛ ПОРТЫ GITLENS QUERY RESULTS (PREVIEW)
+ bash - docker-hive

098cf7dd1eea bde2020/hive:2.3.2-postgresql-metastore "entrypoint.sh /bin/..." 17 seconds ago Up 15 seconds
d24432e90c63 bde2020/hive:2.3.2-postgresql-metastore "entrypoint.sh /opt/..." 17 seconds ago Up 15 seconds
f43207fb3209 bde2020/hadoop-namenode:2.0.0-hadoop2.7.4-java8 "entrypoint.sh /run/..." 17 seconds ago Up 15 seconds (hea
lth: starting) 0.0.0.0:50070->50070/tcp namenode
4324ab4adb8b shawnzhu/prestodb:0.181 "/bin/launcher run" 17 seconds ago Up 15 seconds
730dc23fe7ae bde2020/hadoop-datanode:2.0.0-hadoop2.7.4-java8 "entrypoint.sh /run/..." 17 seconds ago Up 15 seconds (hea
lth: starting) 0.0.0.0:50075->50075/tcp datanode
c5ca1bb13c0e bde2020/hive-metastore-postgresql:2.3.0 "/docker-entrypoint..." 17 seconds ago Up 15 seconds
5432/tcp hive-metastore-postgresql

fred1@DESKTOP-EMH98NN MINGW64 /s/Big_DATA/Big_Data/Big_Data_DZ/task02/docker-hive (master)
$ docker cp AB_NYC_2019.csv namenode:/tmp/
CreateFile S:\Big_DATA\Big_Data\Big_Data_DZ\task02\docker-hive\AB_NYC_2019.csv: The system cannot find the file specified.

fred1@DESKTOP-EMH98NN MINGW64 /s/Big_DATA/Big_Data/Big_Data_DZ/task02/docker-hive (master)
$ docker cp scripts/AB_NYC_2019.csv namenode:/tmp/
Successfully copied 7.08MB to namenode:/tmp/

fred1@DESKTOP-EMH98NN MINGW64 /s/Big_DATA/Big_Data/Big_Data_DZ/task02/docker-hive (master)
$ docker exec -it namenode bash
root@f43207fb3209:/# hdfs dfs -mkdir -p /user/hive/warehouse/airbnb
root@f43207fb3209:/# hdfs dfs -put /tmp/AB_NYC_2019.csv /user/hive/warehouse/airbnb/
root@f43207fb3209:/# exit
exit

fred1@DESKTOP-EMH98NN MINGW64 /s/Big_DATA/Big_Data/Big_Data_DZ/task02/docker-hive (master)
$
```

```
Файл  Правка  Выделение  ...  task02
docker-compose.yml M  Steps.md 9+, U X

docker-hive > Steps.md
26 0. Hive: рассчитать среднее и дисперсию:
27 * 1. docker exec -it hive bash
28 * 2. docker exec -it docker-hive-hive-server-1 bash
29 * 1. beeline -u jdbc:hive2://localhost:10000
30 * 2. hive
31
32 * ```sql

ПРОБЛЕМЫ 13  ВЫХОДНЫЕ ДАННЫЕ  ТЕРМИНАЛ  ПОРТЫ  GITLENS  QUERY RESULTS (PREVIEW)  bash - docker-hive

fred1@DESKTOP-EMH98NN MINGW64 /s/Big_DATA/Big_Data/Big_Data_DZ/task02/docker-hive (master)
$ docker cp AB_NYC_2019.csv namenode:/tmp/
CreateFile S:\Big_DATA\Big_Data\Big_Data_DZ\task02\docker-hive\AB_NYC_2019.csv: The system cannot find the file specified.

fred1@DESKTOP-EMH98NN MINGW64 /s/Big_DATA/Big_Data/Big_Data_DZ/task02/docker-hive (master)
$ docker cp scripts/AB_NYC_2019.csv namenode:/tmp/
Successfully copied 7.08MB to namenode:/tmp/

fred1@DESKTOP-EMH98NN MINGW64 /s/Big_DATA/Big_Data/Big_Data_DZ/task02/docker-hive (master)
$ docker exec -it namenode bash
root@f43207fb3209:/# hdfs dfs -mkdir -p /user/hive/warehouse/airbnb
root@f43207fb3209:/# hdfs dfs -put /tmp/AB_NYC_2019.csv /user/hive/warehouse/airbnb/
root@f43207fb3209:/# exit
exit

fred1@DESKTOP-EMH98NN MINGW64 /s/Big_DATA/Big_Data/Big_Data_DZ/task02/docker-hive (master)
$ docker exec -it hive-server bash
root@098cf7dd1eea:/opt# hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/hive/lib/log4j-slf4j-impl-2.6.2.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop-2.7.4/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]

Logging initialized using configuration in file:/opt/hive/conf/hive-log4j2.properties Async: true
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
hive>
```

```
Файл  Правка  Выделение  ...  task02
docker-compose.yml M  Steps.md 9+, U X

docker-hive > Steps.md
32 * ```sql
33 CREATE EXTERNAL TABLE airbnb (
34     id INT,
35     name STRING,
36     host_id INT,
37     host_name STRING,
38     neighbourhood_group STRING,
39     neighbourhood STRING,
40     latitude DOUBLE,
41     longitude DOUBLE,
42     room_type STRING,
43     price INT,
44     minimum_nights INT,
45     number_of_reviews INT,
46     last_review STRING,
47     reviews_per_month DOUBLE,
48     calculated_host_listings_count INT,
49     availability_365 INT
50 )
51 ROW FORMAT DELIMITED
52 FIELDS TERMINATED BY ','
53 STORED AS TEXTFILE
54 TBLPROPERTIES ("skip.header.line.count"="1");
55
56 LOAD DATA INPATH '/user/hive/warehouse/airbnb/AB_NYC_2019.csv' INTO TABLE airbnb;
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ПРОБЛЕМЫ 13  ВЫХОДНЫЕ ДАННЫЕ  ТЕРМИНАЛ  ПОРТЫ  GITLENS  QUERY RESULTS (PREVIEW)  bash - docker-hive

OK
Time taken: 1.544 seconds
hive>
```

```
Файл  Правка  Выделение  ...  task02
docker-compose.yml M  Steps.md 9+, U X
docker-hive > Steps.md
73  ... sql
75  ... ROUND(AVG(price), 2) AS mean_price,
76  ... ROUND(VARIANCE(price), 2) AS variance_price
77  ... FROM airbnb;
78  ...

ПРОБЛЕМЫ 13  ВЫХОДНЫЕ ДАННЫЕ  ТЕРМИНАЛ  ПОРТЫ  GITLENS  QUERY RESULTS (PREVIEW)
Time taken: 1.544 seconds
OK
OK
Time taken: 1.544 seconds
hive> SELECT
>     ROUND(AVG(price), 2) AS mean_price,
>     ROUND(VARIANCE(price), 2) AS variance_price
>     FROM airbnb;
WARNING: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execu
tion engine (i.e. spark, tez) or using Hive 1.X releases.
Query ID = root_20250419215714_2d934ca1-4cb9-479a-8c2f-203d1f903809
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Job running in-process (local Hadoop)
2025-04-19 21:57:18,591 Stage-1 map = 100%,  reduce = 100%
Ended Job = job_local720961607_0001
MapReduce Jobs Launched:
Stage-Stage-1:  HDFS Read: 28344680 HDFS Write: 0 SUCCESS
Total MapReduce CPU Time Spent: 0 msec
OK
145.12  58528.67
Time taken: 3.707 seconds, Fetched: 1 row(s)
hive> 
```

```
Файл  Правка  Выделение  ...  task02
docker-compose.yml M  Steps.md 9+, U X
docker-hive > Steps.md
116  7.2 Занялся MapReduce(вариант 2):
117
118  * docker exec -it python bash
119  * python script.py
120  * cat AB_NYC_2019.csv | python mapper.py | sort | python reducer.py
121  * python mapper.py < AB_NYC_2019.csv | sort | python reducer.py
122
123  ... docker-compose.yaml
124

ПРОБЛЕМЫ 13  ВЫХОДНЫЕ ДАННЫЕ  ТЕРМИНАЛ  ПОРТЫ  GITLENS  QUERY RESULTS (PREVIEW)
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Job running in-process (local Hadoop)
2025-04-19 21:57:18,591 Stage-1 map = 100%,  reduce = 100%
Ended Job = job_local720961607_0001
MapReduce Jobs Launched:
Stage-Stage-1:  HDFS Read: 28344680 HDFS Write: 0 SUCCESS
Total MapReduce CPU Time Spent: 0 msec
OK
145.12  58528.67
Time taken: 3.707 seconds, Fetched: 1 row(s)
hive> exit;
root@098cf7dd1eea:/opt# exit
exit

fred1@DESKTOP-EMH98NN MINGW64 /s/Big_DATA/Big_Data/Big_Data_DZ/task02/docker-hive (master)
o $ docker exec -it python bash
root@21a367c4a9b7:/scripts# ls
AB_NYC_2019.csv  mapper.py  reducer.py
root@21a367c4a9b7:/scripts# cat AB_NYC_2019.csv | python mapper.py | sort | python reducer.py
Mean: 152.72
Variance: 57672.85
root@21a367c4a9b7:/scripts# 
```

